



Department Of Computer Science and Engineering

VIGNAN'S LARA INSTITUTE OF TECHNOLOGY AND SCIENCE

PROJECT REVIEW
ON

MTCNN TECHNOLOGY FOR REUNIFICATION WITH REWARD OUTCOMES

Under The Esteemed Guidance of
Dr.A.Hanumaiah Ph.D
Professor
CSE Department

Presented By

G.Bhavya Sree	22FE1A0564
K.Purnima	22FE1A0579
M.Anurekha	22FE1A05B5
K.Jaya Krishna	22FE1A0585



Contents

Title

Abstract

Introduction

Existing System

Proposed System

System Architecture

Proposed Methodology

Literature Survey

Conclusion





Abstract

MTCNN Technology For Reunification With Reward Outcomes is a system that helps locate missing individuals using face recognition technology. Users can upload missing person details and photos, which are then displayed on the website for visibility. Public users can upload or scan suspected images to check for possible matches with the database. The system also offers multi-language support, search filters, messaging, community interaction, and rewards, encouraging participation. This makes the search for missing persons faster, more accurate, and collaborative.





Introduction

- Many people (children, elderly, vulnerable) go missing each year.
- Traditional methods (police, posters, media) are slow & limited.
- MTCNN-based face recognition makes searches faster & more accurate.
- Users can upload missing person image & details
- Public users can upload photos /scan face that are matched with existing records for results.





Introduction

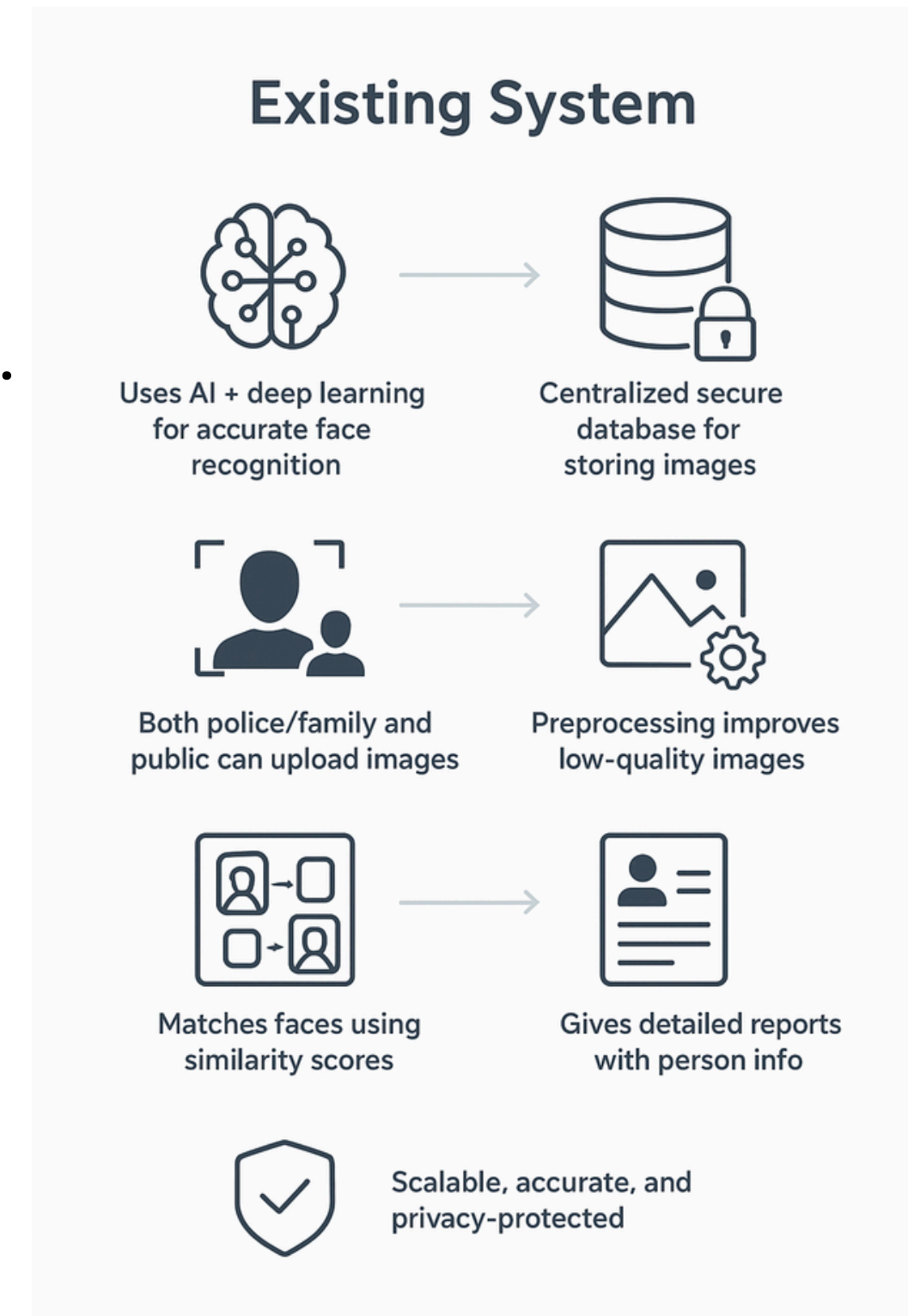
- Supports multiple languages & smart filters (location, reward, etc.).
- Provides messaging, community support & rewards to boost participation.
- NGO/community donations distributed to top 50 high-reward cases.
- Smart, reliable & user-friendly system to reunite missing persons with families.





Existing System

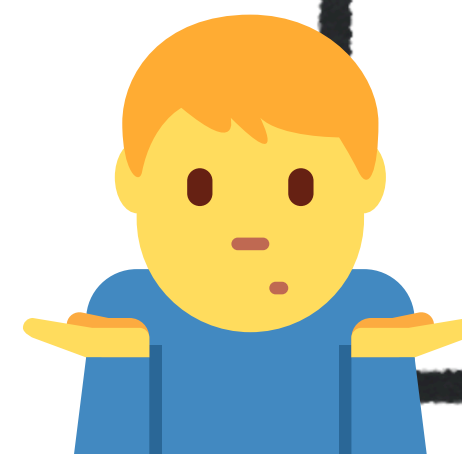
- Uses CNN for accurate face recognition.
- Centralized secure database for storing images.
- Both family and public can upload images.
- Preprocessing improves low-quality images.
- Matches faces using similarity scores.
- Gives detailed reports with person info.
- Scalable, accurate, and privacy-protected.



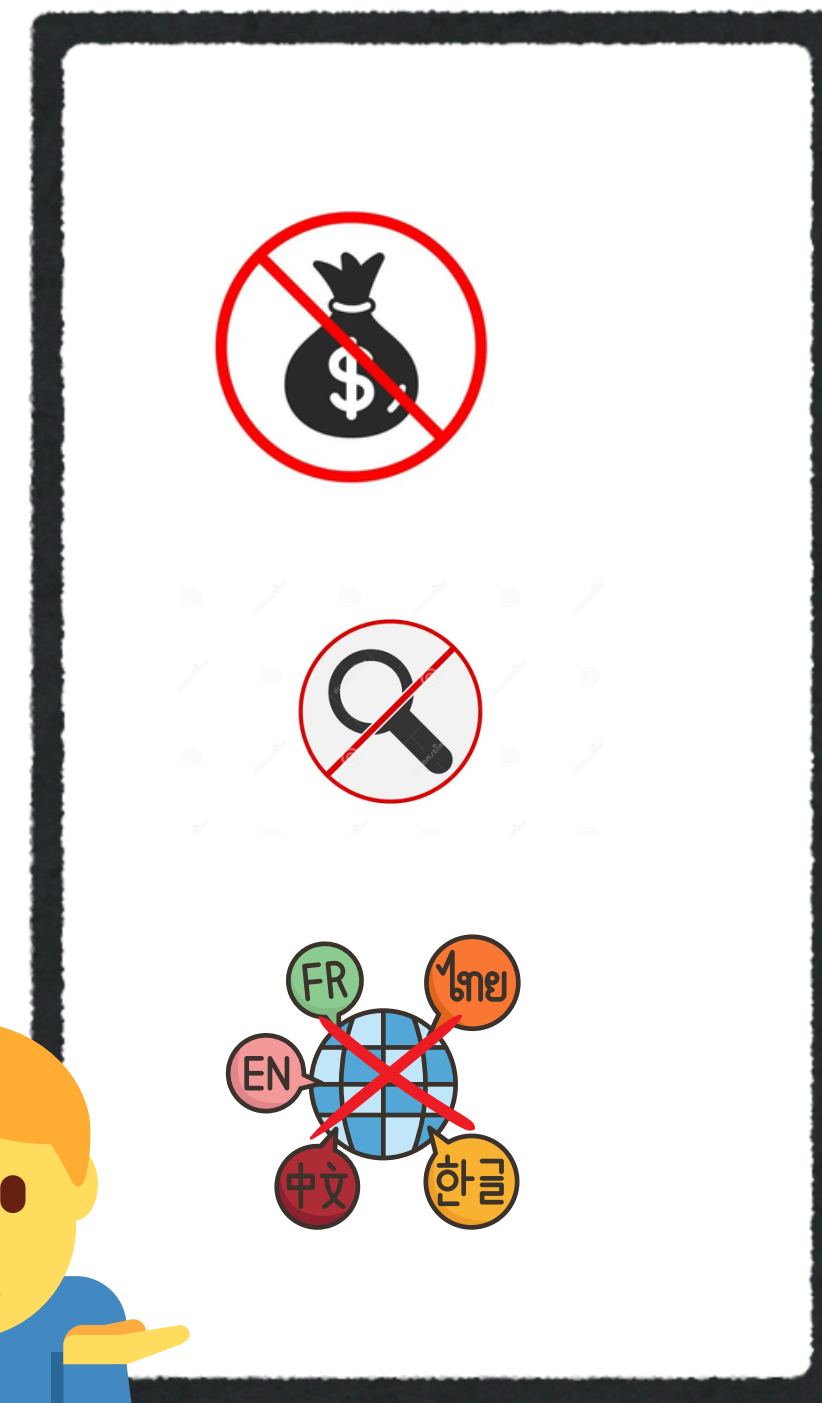


Existing System DrawBacks

- No reward , people may not be interested to help.
- Searching is not proper if data is too much or not managed well.
- No multi-language translator , people from different regions may face difficulties.
- No chat feature,If public user saw in particular location but got missed the case they can unable to contact them



Why I have to Search ?





Proposed System

- **Any One Can upload:**
 - Anyone can add missing person images into the database.
 - Each upload includes description (name, age, last seen, etc.) and a reward amount.
- **When a Public User Identifies or Suspects:**
 - Users can upload images to check for matches (not stored in DB).
 - They can filter cases by location and reward to search in their nearby area.
 - If their upload matches, a report is automatically generated and sent to police, while the user gets a notification.





Proposed System

- **NGO Support System:**

- NGOs can contribute funds to support the reward system.
- Collected reward money can be distributed to the top 50 verified uploads (helpful users).
- This motivates more people to actively participate.



- **Multi-language translator**

- The system supports multiple languages so people from different regions can use it easily.

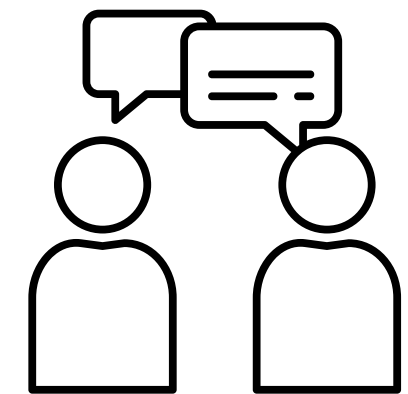




Proposed System

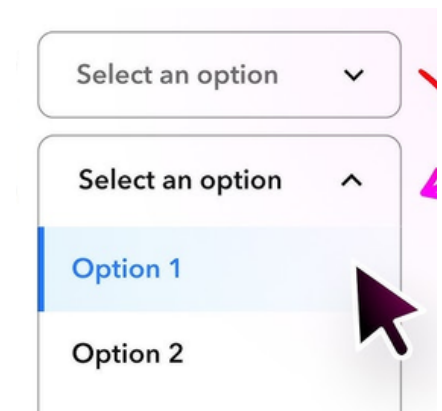
- **Messaging:**

- Users can send messages to each other.
- Enables quick communication for tips, clarifications, or updates.
- Helps speed up investigations with community input.



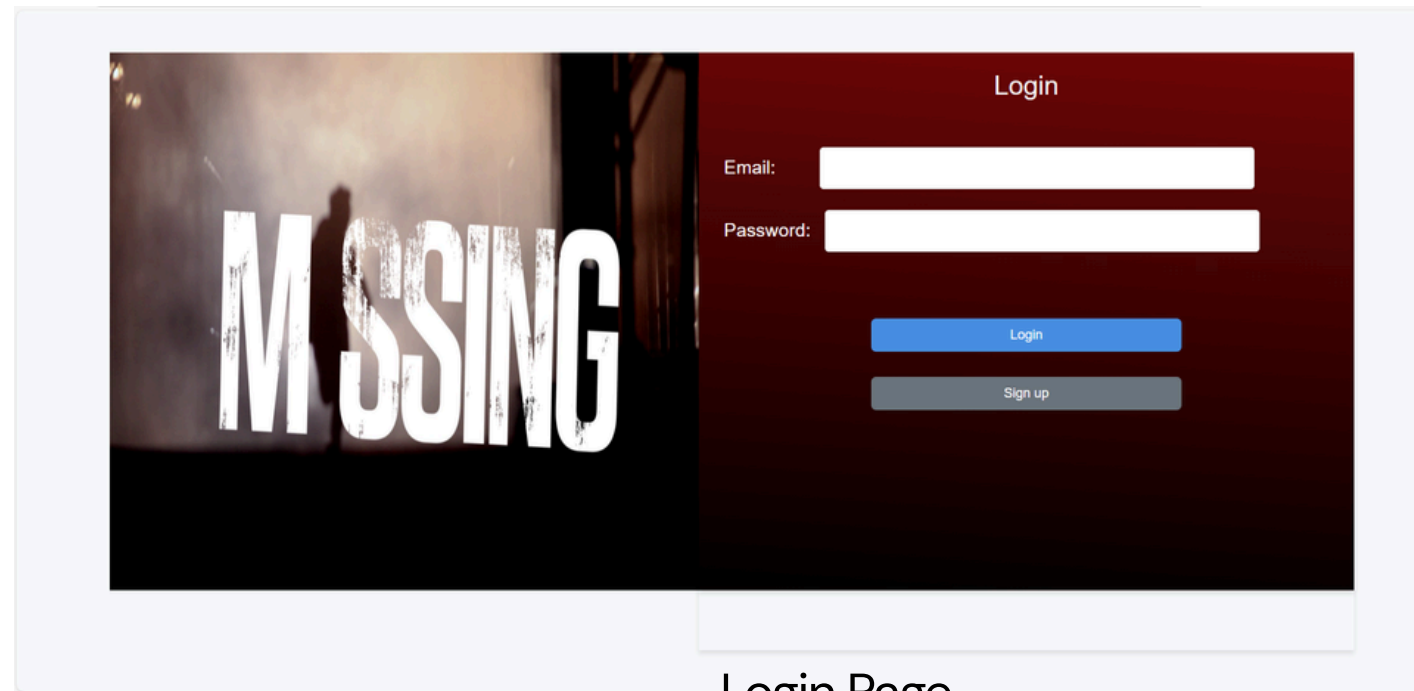
- **Search by Filter:**

- Users can search missing cases by location, reward, identity
- Makes finding relevant cases in their area easier.
- Improves accuracy and saves time compared to manual browsing.

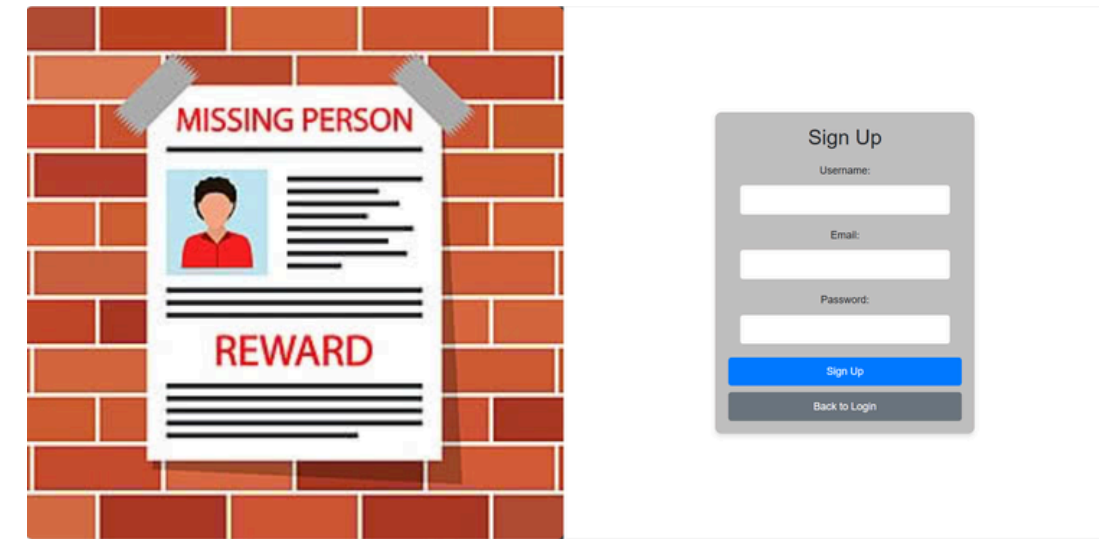




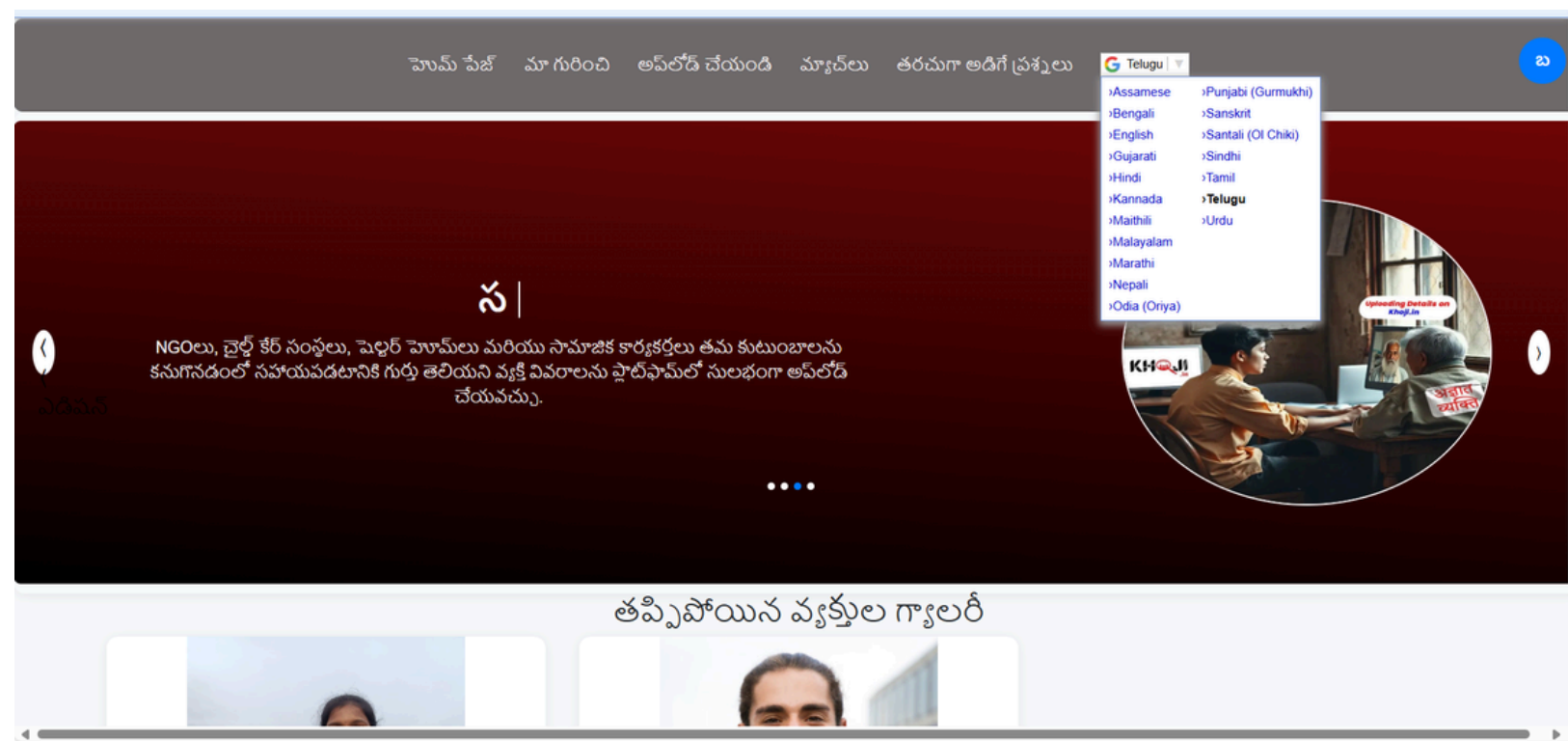
Proposed System



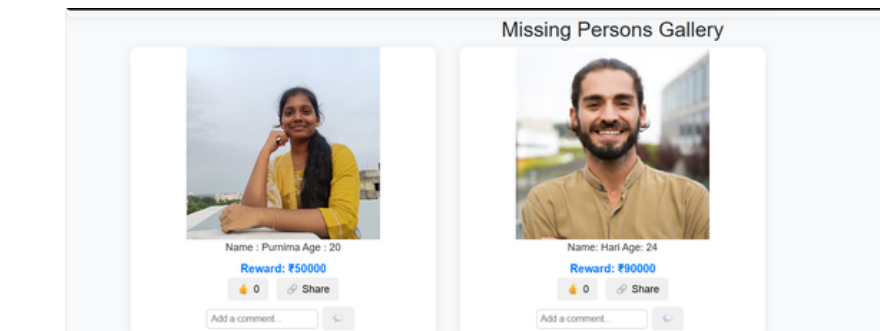
Login Page



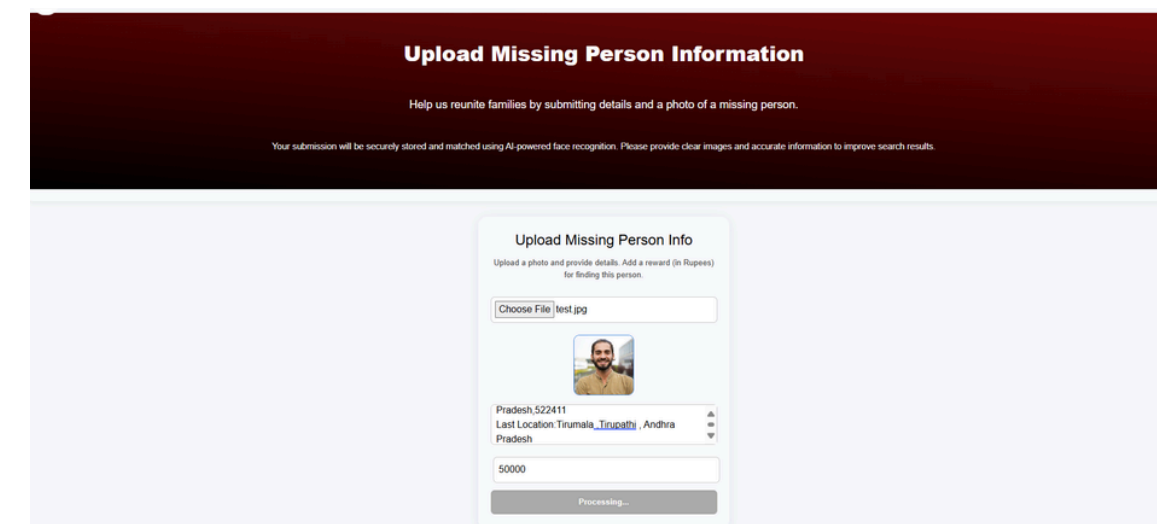
SignUp Page



Multi-Lang-Translator



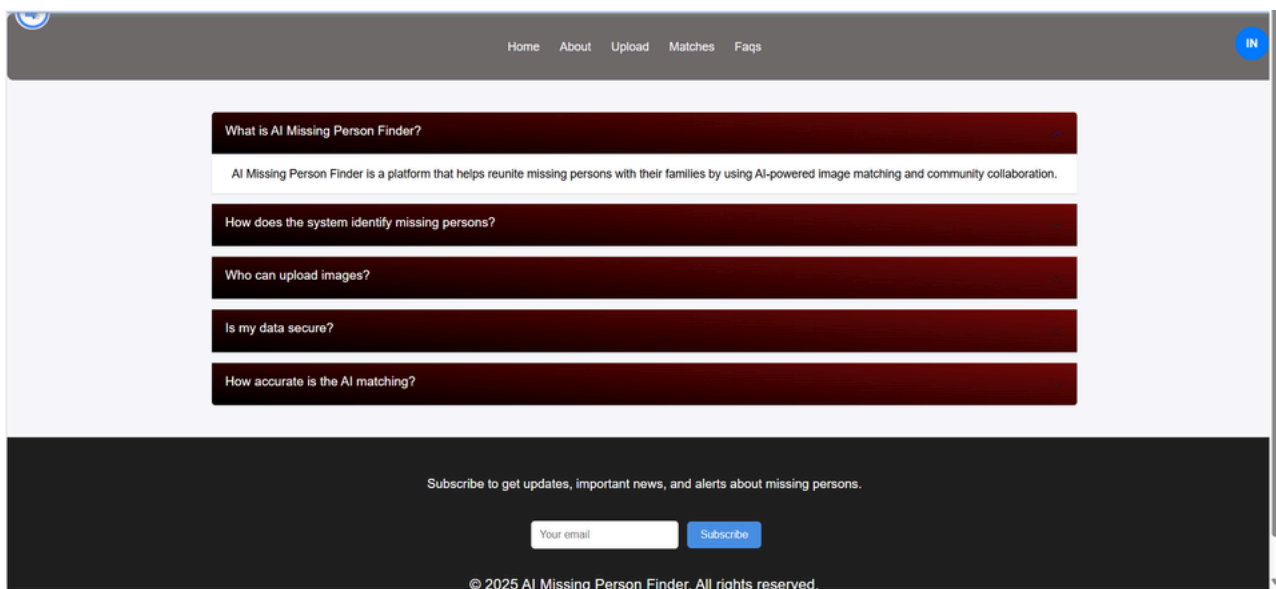
Missing-Galary



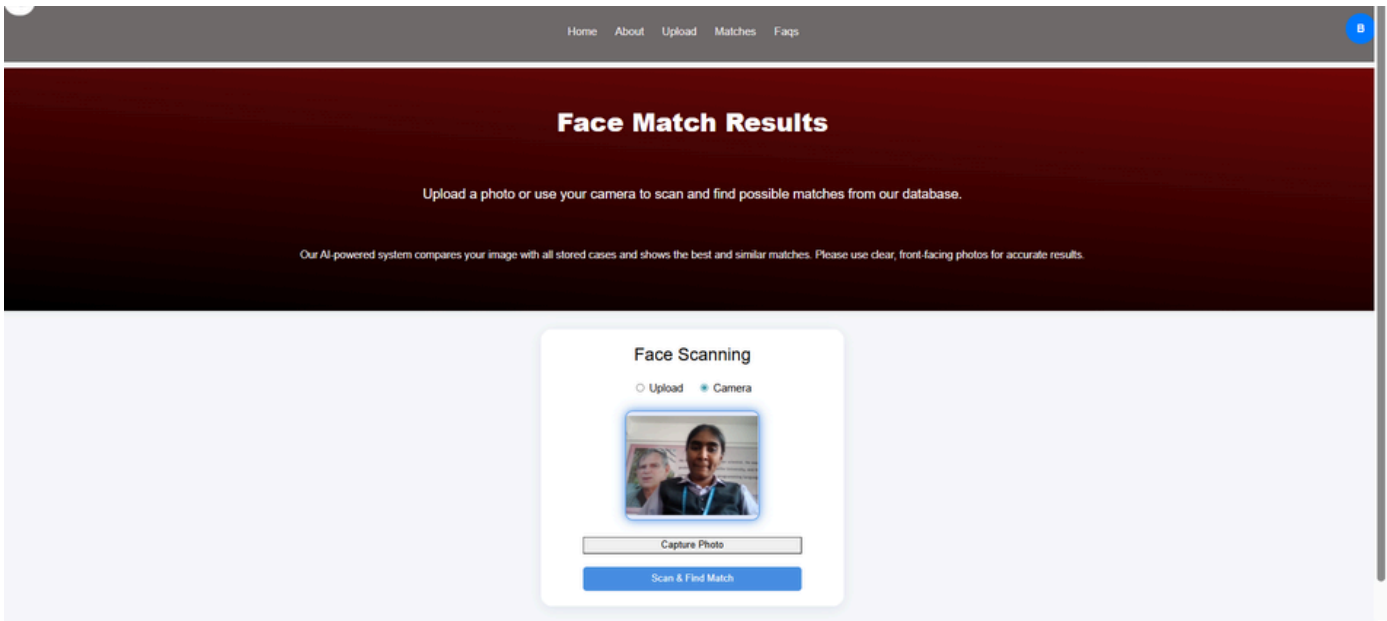
Upload Missing person



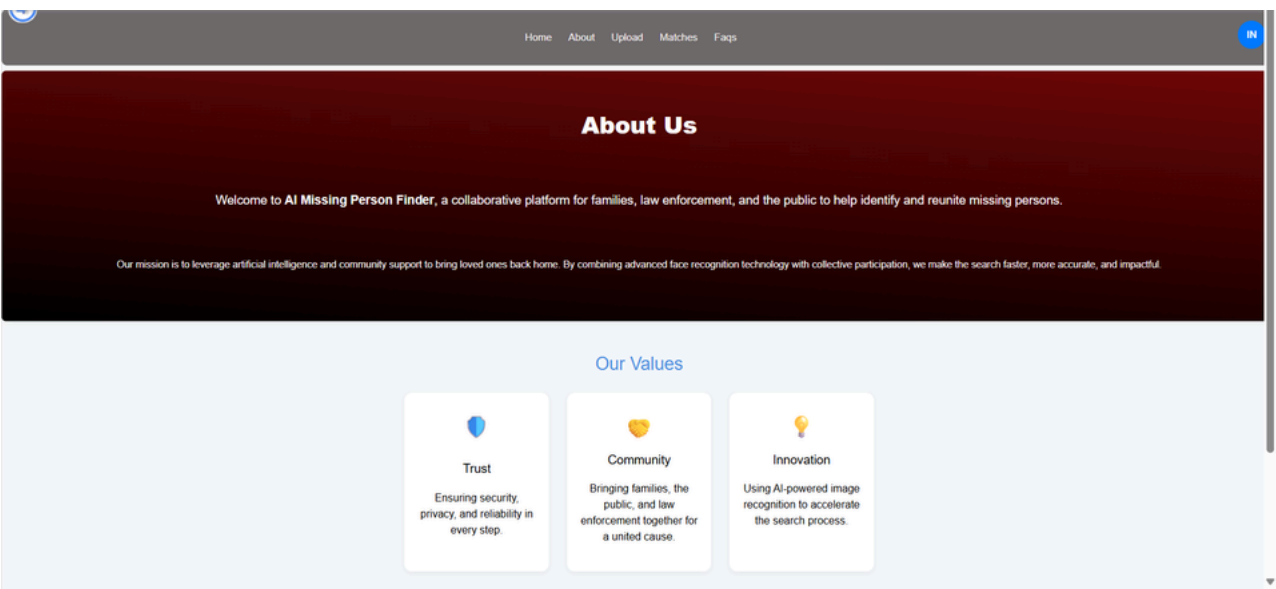
Proposed System



Faqs



CAMERA Scanning



About Us

Face Scanning

☒ Upload ☐ Camera

Choose File test2.webp



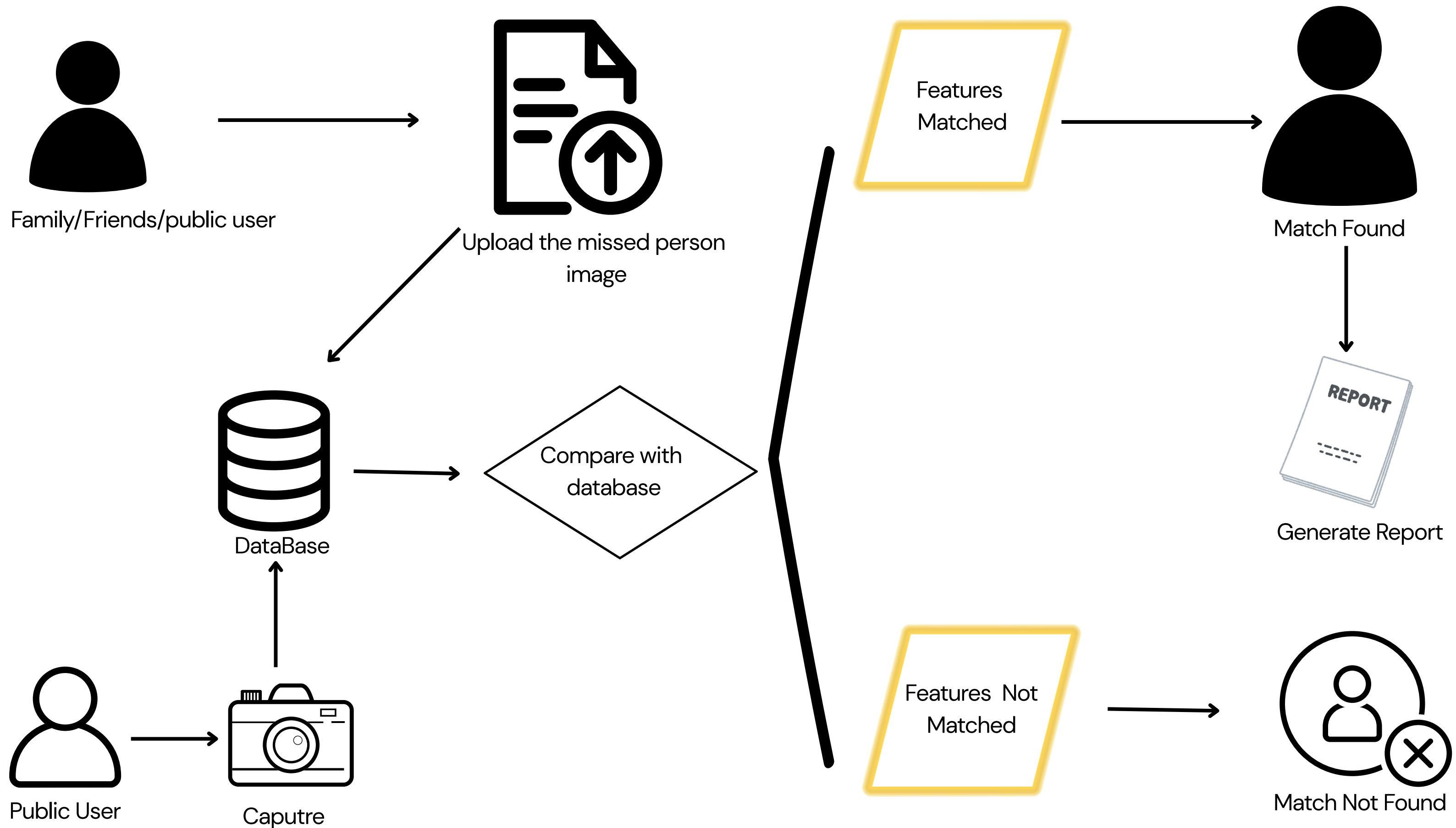
Scan & Find Match

No match found in database.

Upload to match



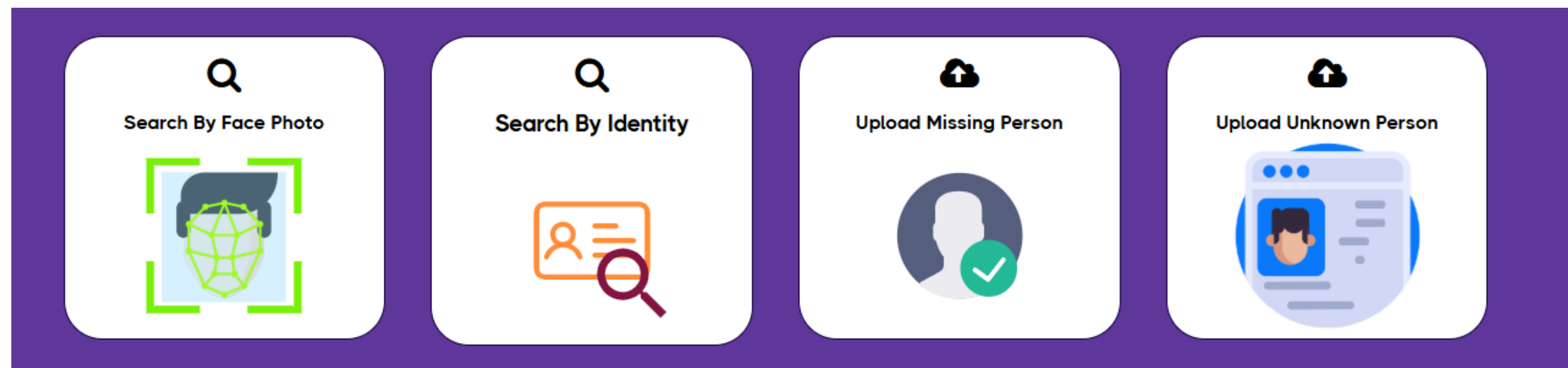
System Architecture





Proposed Methodology

- Users can sign up and log in to access the system.
- Family/friends/users all are allowed to upload missing person details (photo, name, description, last seen location, reward).
- Uploaded details are stored securely in a central database.
- The home page displays missing person cases
- Users can filter or search cases by location and reward.



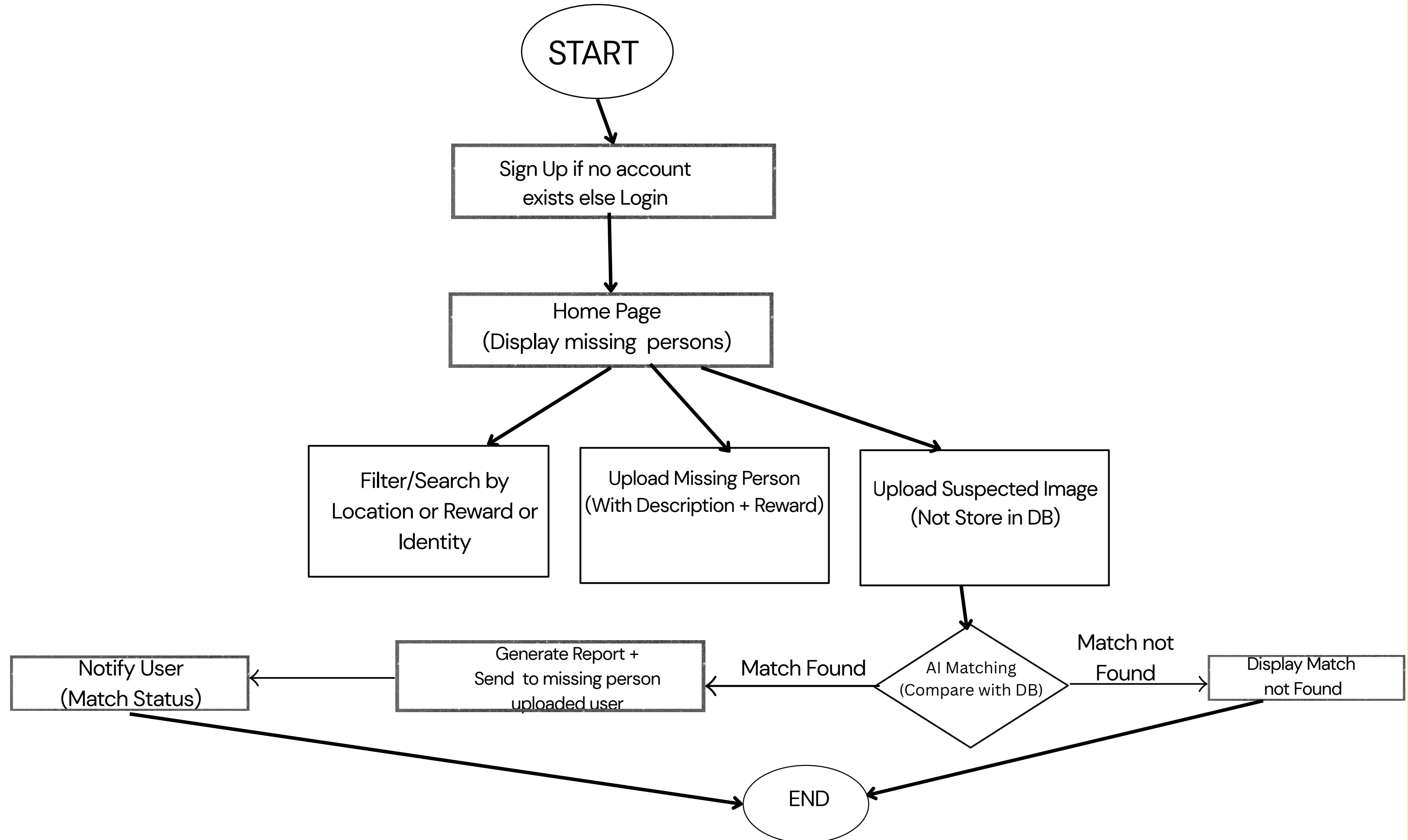


Proposed Methodology

- publicUsers can upload a photo or scan the face of suspicious person for verification (not stored).
- AI compares it with police database images.
- If match → report generated, sent to police, user notified.
- If no match → query discarded.
- Ensures authenticity, privacy, and accurate results.



Proposed Methodology





Literature Survey

S.No	Paper Title	Methodology	Drawbacks	Proposed Methodology to Overcome Drawbacks
1.	High Performance and Efficient Real-time Face Detector on CPU Based on CNN (2020)	Cascaded P-RBM (Probabilistic Restricted Boltzmann Machine) with CNN layers for recognition	Training and implementation is difficult	Use lightweight CNNs, dropout models





Literature Survey

S.No	Paper Title	Methodology	Drawbacks	Proposed Methodology to Overcome Drawbacks
2.	CIS: An Automated Criminal Identification System (2021)	CNN classifier with Keras & TensorFlow, trained on IMDb dataset; Dlib & OpenCV for facial landmarks	Hard to implement and requires large-scale cloud training	Use optimized CNN models with cloud-based distributed training





Literature Survey

S.No	Paper Title	Methodology	Drawbacks	Proposed Methodology to Overcome Drawbacks
3.	Person Re-Identification by Using DPM and CNN-SPP with SVM (2020)	Deformable Part Model + CNN (SPP-Net structure) for feature extraction, SVM for classification	Requires longer time to train and heavy computation	Employ faster DL models (e.g., ResNet, YOLO) with GPU acceleration





Literature Survey

S.No	Paper Title	Methodology	Drawbacks	Proposed Methodology to Overcome Drawbacks
4.	Efficient Face Recognition System for Identifying Lost People (2019)	HOG for initial detection, Deep CNN for feature embeddings, Linear SVM for classification	Accuracy drops significantly for children aged 0–9	Use age-invariant DL models and fine-tuned CNN embeddings





Conclusion

The system aims to reunite missing persons with their loved ones by leveraging MTCNN face recognition to make searches faster and more accurate. It allows users to report cases, verify identities, and communicate within the platform, ensuring a collaborative approach. With additional features like multilingual support, search filters, and rewards, the system enhances transparency, efficiency, and impact in the search for missing people.





Thank You

For your attention

